

Question

The element **A** exists as a gray simple substance with relatively active chemical properties. Mixing the simple substance of **A** with a binary compound **B** containing 35.77% of **A** in a sealed tube at a certain stoichiometric ratio and heating to 393 K yields a brownish-red oily liquid **C** containing 69.01% of **A**; **B** can be prepared by passing Cl_2 into a solution of **C** in CCl_4 . Anhydrous liquid ammonia can react with **C** to produce the simple substance of **A**, while an ammonia solution in diethyl ether reacts with **C** at 193 K to produce an explosive substance **D** with an **A** content of 84.93%. **E**, which is isoelectronic with **C** in a broader sense, can also react with NH_3 to produce a similarly explosive **F** with a nitrogen content of 30.40%.

Reacting **D** with **B** in SOCl_2 at 313 K yields a yellow precipitate. After washing and recrystallization, a yellow compound **G** is obtained. Its cation contains only one **A** and the **A** content in the cation is 36.35%, and the nitrogen content is 19.35%; the overall **A** content in the entire compound is 32.62%.

Which of the following statements is correct:

- A.** All other options are incorrect
- B.** **A** atomic ground state only has one single electron
- C.** **B** and **E** chemical formulas have the same number of atoms.
- D.** **B** and **D** have the same number of atoms in their chemical formulas.
- E.** **D** and **F** have different numbers of atoms in their chemical formulas.
- F.** **G** is a 1:1 type ionic compound
- G.** **G** anion possesses a tetrahedral configuration.

H. The chemical formula of **G** contains 18 atoms after removing atom **A**.

Answer

Correct Answer: **H**

Detailed Explanation

Cl_2 is passed into **C** to obtain a binary compound **B**. It can be determined that **B** is a chloride; assume the chemical formula of **B** is ACl_n , calculate based on the mass fraction:

CHECKPOINT

1 PTS

B is a chloride

If n is 1, the relative atomic mass of **A** is $35.45 / (1 - 0.3577) - 35.45 = 19.7$, no reasonable element.

If n is 2, the relative atomic mass of **A** is $35.45 \times 2 / (1 - 0.3577) - 35.45 \times 2 = 39.5$, no reasonable element.

If n is 3, the relative atomic mass of **A** is $35.45 \times 3 / (1 - 0.3577) - 35.45 \times 3 = 59.2$, no reasonable element.

If n is 4, the relative atomic mass of **A** is $35.45 \times 4 / (1 - 0.3577) - 35.45 \times 4 = 78.9$, consider it to be Se, the valence state is reasonable.

No reasonable element exists when n is greater than or equal to 5.

In summary, it is determined that **A** is Se, then **B** is SeCl_4 . Se has two unpaired electrons, so option B is incorrect.

CHECKPOINT

2 PTS

A is Se

CHECKPOINT

1 PTS

B is SeCl_4

The content of Se in **C** is 69.01% and should only contain Se, Cl, it can be deduced that **C** is Se_2Cl_2 .

CHECKPOINT

1 PTS

C is Se_2Cl_2

The content of Se in **D** is 84.93%; since **D** is an explosive, consider it to be a nitride; according to the mass fraction, it is easy to obtain the simplest chemical formula of **D** as SeN , according to the properties of chalcogen elements, it is known that this compound is Se_4N_4 .

CHECKPOINT

1 PTS

D is an explosive, consider it to be a nitride

CHECKPOINT

1 PTS

The simplest chemical formula of **D** is SeN , actually this compound is Se_4N_4

E is the generalized isoelectronic species of Se_2Cl_2 , then **E** is likely to be S_2Cl_2 or Te_2Cl_2 or the uncommon O_2Cl_2 ; **E** can also generate the explosive substance **F**, with a nitrogen content of 30.40%, calculation yields the simplest chemical formula as SN , actually S_4N_4 ; then **E** is S_2Cl_2 . From this, it can be seen that options C, D, and E are all incorrect.

CHECKPOINT

1 PTS

E is S_2Cl_2

CHECKPOINT

1 PTS

The simplest chemical formula of **F** is SN , actually this compound is S_4N_4

Se_4N_4 and SeCl_4 can synthesize **G**. Its cation contains only one Se; and the Se content is 36.35%, the nitrogen content is 19.35%; based on this, the ratio of the number of Se and N in the cation can be calculated as $78.96/0.3635 : 14.01/0.1935 = 1 : 3$, containing only one Se, and the chemical formula of the cation can be calculated as $\text{SeN}_3\text{S}_3^{n+}$ according to the synthesis method.

CHECKPOINT

1 PTS

In **G** cation, the ratio of the number of Se and N is 1 : 3

CHECKPOINT

1 PTS

The chemical formula of **G** cation is SeN_3S_3

The Se content in the entire compound is 32.62%, which is smaller than the Se content in the cation, meaning that the anion must contain Se; assuming the anion contains 1 Se, **G** contains one cation and n anions, then the relative molecular mass of the anion is $78.96 \times (1 + n)/0.3262 - 78.96/0.3635 = 242n + 25$, subtracting the mass of n Se atoms, when n is 0.5, exactly the relative molecular mass of three chlorines remain; therefore, it is obvious that the anion is SeCl_6^{2-} ; at this time, the cation must be SeN_3S_3^+ to match the mass fraction. Therefore, the chemical formula of **G** is $[\text{SeN}_3\text{S}_3]_2[\text{SeCl}_6]$, option F is incorrect, and H is correct. SeCl_6^{2-} has an octahedral configuration, so option G is incorrect.

CHECKPOINT

2 PTS

G anion is SeCl_6^{2-} , octahedral configuration

CHECKPOINT

1 PTS

G cation is SeN_3S_3^+

CHECKPOINT

2 PTS

The chemical formula of **G** is $[\text{SeN}_3\text{S}_3]_2[\text{SeCl}_6]$, which is a 1:2 type ionic compound